

Podcast 26 — IaC & CI/CD

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Length target: 15 min **Repo references:** `domain-2-new-solutions/21-cloudformation.md`, `domain-4-migration/07-cicd.md`, `domain-2-new-solutions/44-codeguru.md`, `39-proton.md`, `domain-1-organizational/48-service-catalog.md`

Topic & Scope

CloudFormation (and SAM/CDK), CodePipeline + CodeBuild + CodeDeploy + CodeArtifact, CodeGuru, plus the platform tier (Proton, Service Catalog). Deployment strategies + CI/CD patterns.

Service Coverage Depth

Deep: - CloudFormation — templates, change sets, stack sets, drift detection, custom resources, nested stacks - CodePipeline + CodeBuild + CodeDeploy — pipeline orchestration, build, deployment strategies - Deployment strategies (rolling, blue/green, canary, linear)

Brief: - SAM (Lambda-focused IaC) - CDK (code-as-IaC) - CodeArtifact (private package registry) - CodeGuru (reviewer + profiler + security) - Proton + Service Catalog (covered in podcasts 15 + 4)

Structured Outline

1. **Open (30s)** — “IaC + CI/CD = how you deploy reliably. CloudFormation, the Code* family, deployment strategies — let’s go.”
2. **CloudFormation Deep (4 min)** — stacks, templates (YAML/JSON), parameters, mappings, conditions, outputs, change sets, drift detection, custom resources (Lambda-backed), nested stacks, StackSets (multi-account / multi-Region)
3. **SAM + CDK (1.5 min)** — SAM = Lambda + serverless transform; CDK = code (Python/TS/Java/C#/Go) compiles to CloudFormation
4. **CodePipeline + CodeBuild + CodeDeploy (4 min)** — Pipeline = orchestration with stages + actions; Build = managed build containers; Deploy = blue/green for Lambda + ECS, in-place + blue/green for EC2 ASG
5. **Deployment Strategies (2.5 min)** — Rolling (in-place), Blue/Green (two environments swap), Canary (small % then full), Linear (incremental %), shift via deployment configuration
6. **Tooling Around (2 min)** — CodeArtifact (private packages), CodeGuru Reviewer / Profiler / Security, Service Catalog for vending, Proton for platform engineering
7. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- CloudFormation template sections: Parameters, Mappings, Conditions, Resources (required), Outputs, Transform, Metadata
- Change sets preview changes before applying — use for safe updates
- Drift detection identifies manual changes outside CFN
- Custom resources: Lambda-backed; for resources CFN doesn't support natively
- Nested stacks: modularize templates
- StackSets: deploy templates across accounts + Regions (self-managed or service-managed via Organizations)
- DeletionPolicy (Retain / Snapshot / Delete) + UpdateReplacePolicy
- SAM Transform: macro that simplifies Lambda + API Gateway + DynamoDB authoring
- CDK: write IaC in TypeScript / Python / Java / C# / Go; synthesizes to CFN
- CDK constructs (L1/L2/L3) — L2 abstracts CFN; L3 is patterns
- CodePipeline: source → build → test → deploy stages with actions
- CodePipeline V1 vs V2 — V2 supports triggers, variables, parameters
- CodeBuild: managed build containers, custom or AWS-curated, integrates with secrets
- CodeDeploy strategies for Lambda: Canary (e.g., 10% for 5min then 100%), Linear (10% every X min)
- CodeDeploy strategies for ECS: Blue/Green via ALB target group swap
- CodeDeploy strategies for EC2 ASG: in-place rolling or blue/green
- CodeArtifact: private package repository for npm / Maven / PyPI / NuGet / Cargo / Ruby
- CodeGuru Reviewer: ML code review on PRs (Java + Python)
- CodeGuru Profiler: production profiling
- CodeGuru Security: source code security scanning

Must-Mention Exam Traps

- EXAM TRAP: CloudFormation rolls back on failure by default — disable rollback for debugging
- EXAM TRAP: Stack deletion can fail on RetentionPolicy=Retain resources — they stay
- EXAM TRAP: Change sets don't auto-apply — must execute
- EXAM TRAP: StackSet service-managed needs Organizations + delegated administrator
- EXAM TRAP: CodeCommit is closed to new customers — use GitHub / GitLab / Bitbucket / CodeConnections
- EXAM TRAP: CodeDeploy in-place takes downtime; blue/green doesn't
- EXAM TRAP: Lambda Canary requires alias + traffic shifting — not Linear deployment by version directly
- EXAM TRAP: CodeDeploy blue/green for EC2 needs a separate ASG for green — uses ELB target group swap
- EXAM TRAP: CodeBuild buildspec.yml location — at root by default; configurable
- EXAM TRAP: CodeDeploy hooks (BeforeInstall, AfterInstall, etc.) for EC2 require agent
- EXAM TRAP: CDK ≠ CloudFormation directly — CDK synthesizes to CFN, but operates via CFN under the hood

- EXAM TRAP: SAM is for serverless — for general infra use CFN/CDK
- EXAM TRAP: Service Catalog ≠ CodePipeline — Catalog is curated vending; Pipeline is CI/CD
- EXAM TRAP: Proton is platform engineering — not a CI/CD tool itself (uses CodePipeline under the hood)
- EXAM TRAP: Blue/Green uses TWO full environments — costs double during deploy
- EXAM TRAP: CodeBuild compute types: Lambda compute available for short builds (cheaper); managed EC2 for long

Key Decision Matrix

Scenario	Pick
YAML IaC	CloudFormation
Code IaC (TypeScript, Python)	CDK
Lambda-focused IaC	SAM
Multi-account multi-Region template deploy	StackSets
Pipeline orchestration source-to-deploy	CodePipeline
Managed build environment	CodeBuild
Blue/green for ECS / Lambda	CodeDeploy
Private npm / PyPI packages	CodeArtifact
Automated PR code review	CodeGuru Reviewer
Production performance profiling	CodeGuru Profiler
Source code security scan	CodeGuru Security
Vending compliant templates to dev teams	Service Catalog
Platform engineering with versioned env + service templates	Proton
Zero-downtime deployment	Blue/Green or Canary
Incremental rollout by percentage	Linear deployment

Tone & Style

- Lead with CloudFormation (most-tested)
- For deployment strategies, walk through with concrete percentages
- Mention CDK as the modern IaC choice without abandoning CFN underneath

Rapid-Fire Closer

“YAML IaC?” — “CloudFormation.” “Code IaC?” — “CDK.” “Multi-account templates?” — “StackSets.” “Lambda blue/green?” — “CodeDeploy with alias + traffic shift.” “Private npm?” — “CodeArtifact.” “ML code review?” — “CodeGuru Reviewer.”